`:00198 supports` — audit tables

Generated from the current ROADMAP ontology. `:00198` is **not** transitive and has no property chain, so nothing here is inferred: every link shown is asserted.

Code	Principle	Framework
F	findability	FAIR
A	accessibility	FAIR
I	interoperability	FAIR
R	reusability	FAIR
Un	universality	FUTURE-AI
Tr	traceability	FUTURE-AI
Ex	explainability	FUTURE-AI
Us	usability	FUTURE-AI
Ro	robustness	FUTURE-AI
Fa	fairness	FUTURE-AI

Table 1 — Metadata elements with a *direct* `supports` link to a principle

ID	Metadata element	Principles	Also supports *n* guideline items
`:03788`	AI resource metadata	Tr	0
`:00051`	annotation level	Ro	13
`:00069`	contact information	A R Us	3
`:03351`	decision threshold	Ex Ro	5
`:03614`	degree of automation	Us	6
`:00068`	license	A R Us	10
`:03723`	link	F A	11
`:03598`	model architecture	I R	11
`:03373`	model availability	A R Us	10
`:03481`	model input	I	19
`:03482`	model output	I	10
`:03808`	model performance	Ro	16
`:00011`	model use	Us	13
`:00036`	model user	Us	9
`:03597`	name	F	1
`:03411`	noise	Ro	6
`:00067`	reference	R	6
`:03602`	regulatory information	Ro	1
`:03620`	reproducibility	Ro	7
`:03396`	sampling	Un	27
`:00167`	summary	F R Ex Us	5
`:00066`	version	F R Ro	7

22 metadata classes carry **36** direct principle links. The remaining 172 metadata classes reach principles only indirectly, through guideline items, or not at all.

Table 2 — Guideline items appearing as subject or object of `supports`

Column 3 lists metadata elements that support the item. Column 4 lists principles the item supports.

Guideline item	Definition	Metadata elements	Principles
CLAIM item 01 `:00203`	Identification as a study of AI methodology, specifying the category of technology used (e.g., deep learning).	model architecture	F
CLAIM item 02 `:00204`	Summary of study design, methods, results, and conclusions.	—	F Us
CLAIM item 03 `:00205`	Scientific and/or clinical background, including the intended use and role of the AI approach.	annotation level, annotator, clinical benefit, clinical workflow phase, indication for use, instructions for use, model use, model user, summary	R Un Ex Us
CLAIM item 04 `:00206`	Study aims, objectives, and hypotheses.	motivation	F Un Us
CLAIM item 05 `:00207`	Prospective or retrospective study.	—	Un Ro
CLAIM item 06 `:00208`	Study goal.	motivation	Un Us Ro
CLAIM item 07 `:00209`	Data sources.	model input, sampling	F A I R Un
CLAIM item 08 `:00210`	Inclusion and exclusion criteria.	sampling	R Un Ro Fa
CLAIM item 09 `:00211`	Data pre-processing.	pre-processing	I R Ro
CLAIM item 10 `:00212`	Selection of data subsets	partitioning scheme, sampling, subset criterion	R Ro Fa
CLAIM item 11 `:00213`	De-identification methods.	Protected Health Information, confidentiality, risk of re-identification, sensitive data	R Ro
CLAIM item 12 `:00214`	How missing data were handled.	missing information	R Un Ro
CLAIM item 13 `:00215`	Image acquisition protocol.	image characteristic, image file format	I R Un Ro
CLAIM item 14 `:00216`	Definition of method(s) used to obtain reference standard.	annotation level, annotator	I R Ex Ro
CLAIM item 15 `:00217`	Rationale for choosing the reference standard.	—	R Ex Ro

Guideline item	Definition	Metadata elements	Principles
CLAIM item 16 `:00218`	Source of reference standard annotations.	annotation level, annotator	I R Ex Ro
CLAIM item 17 `:00219`	Annotation of test set.	annotation level, annotator	R Ro
CLAIM item 18 `:00221`	Measures of inter- and intra-rater variability of features described by the annotators.	annotator	Ex Ro
CLAIM item 19 `:00220`	How data were assigned to partitions.	partitioning scheme, relationships between instances	R Ro
CLAIM item 20 `:00223`	Level at which partitions are disjoint.	partitioning scheme, relationships between instances	R Ro
CLAIM item 21 `:00224`	Intended sample size.	count, sampling	Ro
CLAIM item 22 `:00225`	Detailed description of model.	decision threshold, degree of automation, model architecture, model input, model output	I R Ex Us Ro
CLAIM item 23 `:00226`	Software libraries, frameworks, and packages.	—	I R Us Ro
CLAIM item 24 `:00227`	Initialization of model parameters.	—	Ro
CLAIM item 25 `:00228`	Details of training approach.	missing information, model input	R Ex Ro
CLAIM item 26 `:00229`	Method of selecting the final model.	—	Un Ex Ro
CLAIM item 27 `:00230`	Ensembling techniques.	—	Ro
CLAIM item 28 `:00231`	Metrics of model performance.	model performance	Un Ex Us Ro Fa
CLAIM item 29 `:00232`	Statistical measures of significance and uncertainty.	model performance, uncertainty	Ro
CLAIM item 30 `:00233`	Robustness or sensitivity analysis.	noise, reproducibility	Un Ex Ro
CLAIM item 31 `:00234`	Methods for explainability or interpretability.	instructions for use, model architecture, model output	Ex Us

Guideline item	Definition	Metadata elements	Principles
CLAIM item 32 `:00235`	Evaluation on internal data.	internal testing	Ro
CLAIM item 33 `:00236`	Testing on external data.	external testing	Un Ro
CLAIM item 34 `:00237`	Clinical trial registration.	—	F R
CLAIM item 35 `:00238`	Numbers of patients or examinations included and excluded.	count, sampling	R Un Fa
CLAIM item 36 `:00239`	Demographic and clinical characteristics of cases in each partition.	count, partitioning scheme, relationships between instances	I R Un
CLAIM item 37 `:00240`	Performance metrics and measures of statistical uncertainty.	model performance, uncertainty	Ex Us Ro Fa
CLAIM item 38 `:00241`	Estimates of diagnostic performance and their precision.	model performance, uncertainty	Ex Ro
CLAIM item 39 `:00242`	Failure analysis of incorrect results.	limitations, model output	Ex Ro Fa
CLAIM item 40 `:00243`	Study limitations.	limitations	Un Ex Ro
CLAIM item 41 `:00244`	Implications for practice, including intended use and/or clinical role.	clinical benefit, clinical workflow phase, decision threshold, indication for use, instructions for use, model use, model user, recommendation	R Un Ex Us Ro
CLAIM item 42 `:00245`	Provide a reference to the full study protocol or to additional technical details.	reference	F A R Un Us
CLAIM item 43 `:00246`	Statement about the availability of software, trained model, and/or data.	license, model availability, reference	F A R Us Ro
CLAIM item 44 `:00247`	Sources of funding and other support; role of funders.	funding	—
CONSORT-AI item 01a `:00332`	Identification as a randomized trial in the title. Indicate that the intervention involves artificial intelligence/machine learning in the title and/or abstract and specify the type of model. State the intended use of the AI intervention	model architecture, model use	F

Guideline item	Definition	Metadata elements	Principles
	within the trial in the title and/or abstract.		
CONSORT-AI item 01b `:00333`	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts). Indicate that the intervention involves artificial intelligence/machine learning in the title and/or abstract and specify the type of model. State the intended use of the AI intervention within the trial in the title and/or abstract.	summary	F
CONSORT-AI item 02a `:00334`	Scientific background and explanation of rationale. Explain the intended use of the AI intervention in the context of the clinical pathway, including its purpose and its intended users (for example, healthcare professionals, patients, public).	model use, model user	R Ex Us
CONSORT-AI item 02b `:00335`	Specific objectives or hypotheses.	motivation	F
CONSORT-AI item 03a `:00336`	Description of trial design (such as parallel, factorial) including allocation ratio.	—	Ro
CONSORT-AI item 04a `:00338`	Eligibility criteria for participants. State the inclusion and exclusion criteria at the level of participants. State the inclusion and exclusion criteria at the level of the input data.	model input, sampling	R Ro Fa
CONSORT-AI item 04b `:00339`	Settings and locations where the data were collected. Describe how the AI intervention was integrated into the trial setting, including any onsite or offsite requirements.	clinical workflow phase	Un Us
CONSORT-AI item 05 `:00340`	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered. State which version of the AI algorithm was used.Describe how the input data were acquired and selected for the AI intervention.Describe how poor quality or unavailable input data were assessed and handled.Specify whether there was human–AI interaction in the handling of the input data, and what level of expertise was required of users.Specify the output of the AI	degree of automation, instructions for use, missing information, model input, model output, model user, version	I R Ex

Guideline item	Definition	Metadata elements	Principles
	interventionExplain how the AI intervention's outputs contributed to decision-making or other elements of clinical practice.		
CONSORT-AI item 06a `:00341`	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed.	—	Ro
CONSORT-AI item 07a `:00343`	How sample size was determined.	—	Ro
CONSORT-AI item 07b `:00344`	When applicable, explanation of any interim analyses and stopping guidelines.	—	Ro
CONSORT-AI item 08a `:00346`	Method used to generate the random allocation sequence.	—	Ro
CONSORT-AI item 08b `:00347`	Type of randomization; details of any restriction (such as blocking and block size).	—	Ro
CONSORT-AI item 09 `:00348`	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned.	—	Ro
CONSORT-AI item 11a `:00350`	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how.	—	Ro
CONSORT-AI item 11b `:00351`	If relevant, description of the similarity of interventions.	—	Ro
CONSORT-AI item 12a `:00352`	Statistical methods used to compare groups for primary and secondary outcomes.	—	Ro
CONSORT-AI item 12b `:00353`	Methods for additional analyses, such as subgroup analyses and adjusted analyses.	—	Ex Fa
CONSORT-AI item 13b `:00355`	For each group, losses and exclusions after randomization, together with reasons.	—	Fa
CONSORT-AI item 14b `:00357`	Why the trial ended or was stopped.	—	Ex

Guideline item	Definition	Metadata elements	Principles
CONSORT-AI item 15 :00358	A table showing baseline demographic and clinical characteristics for each group.	—	R Un Fa
CONSORT-AI item 17a :00360	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval).	model performance	Ex Ro
CONSORT-AI item 17b :00361	For binary outcomes, presentation of both absolute and relative effect sizes is recommended.	—	Ex
CONSORT-AI item 18 :00362	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory.	—	Ex Fa
CONSORT-AI item 19 :00363	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms). Describe results of any analysis of performance errors and how errors were identified, where applicable. If no such analysis was planned or done, justify why not.	known bias or failure mode	Ro Fa
CONSORT-AI item 20 :00364	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses.	limitations	Un Ex
CONSORT-AI item 21 :00365	Generalizability (external validity, applicability) of the trial findings.	limitations	Un Ro
CONSORT-AI item 22 :00366	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence.	clinical benefit	Ex Us
CONSORT-AI item 23 :00367	Registration number and name of trial registry.	clinical trial number	F
CONSORT-AI item 24 :00368	Where the full trial protocol can be accessed, if available.	reference	F A
CONSORT-AI item 25 :00369	Sources of funding and other support (such as supply of drugs), role of funders. State whether and how the AI intervention and/or its code can be accessed, including any restrictions to access or re-use.	funding, license, link, model availability	A

Guideline item	Definition	Metadata elements	Principles
DECIDE-AI item 1 - Title `:00566`	Identify the study as early clinical evaluation of a decision support system based on AI or machine learning, specifying the problem addressed.	—	F
DECIDE-AI item 10 - Implementation `:00581`	a) Report on the user exposure to the AI system, on the number of instances the AI system was used, and on the users' adherence to the intended implementation.	instructions for use	Us Ro
DECIDE-AI item 11 - Modifications `:00584`	Report any changes made to the AI system or its hardware platform during the study. Report the timing of these modifications, the rationale for each, and any changes in outcomes observed after each of them.	update date	Ro
DECIDE-AI item 12 - Human-computer agreement `:00585`	Report on the user agreement with the AI system. Describe any instances of and reasons for user variation from the AI system's recommendations and, if applicable, users changing their mind based on the AI system's recommendations.	degree of automation	Ex Us
DECIDE-AI item 13 - Safety and errors `:00586`	a) List any significant errors/malfunctions related to: AI system recommendations, supporting software/hardware, or users. Include details of: (i) rate of occurrence, (ii) apparent causes, (iii) whether they could be corrected, and (iv) any significant potential impacts on patient care.	known bias or failure mode	Ex Ro Fa
DECIDE-AI item 14 - Human factors `:00587`	a) Report on the usability evaluation, according to recognised standards or frameworks.	—	Ex Us
DECIDE-AI item 15 - Support for intended use `:00588`	Discuss whether the results obtained support the intended use of the AI system in clinical settings.	clinical benefit, model use, recommendation	Ex Us
DECIDE-AI item 16 - Safety and errors `:00589`	Discuss what the results indicate about the safety profile of the AI system. Discuss any observed errors/malfunctions and instances of harm, their implications for patient care, and whether/how they can be mitigated.	known bias or failure mode	Un Ex
DECIDE-AI item 17 - Data	Disclose if and how data and relevant code are available.	license, link, model availability	F A R

Guideline item	Definition	Metadata elements	Principles
availability `:00591`			
DECIDE-AI item 2 - Intended use `:00568`	a) Describe the targeted medical condition(s) and problem(s), including the current standard practice, and the intended patient population(s). b) Describe the intended users of the AI system, its planned integration in the care pathway, and the potential impact, including patient outcomes, it is intended to have.	clinical benefit, clinical workflow phase, indication for use, model use, model user	R Ex Us
DECIDE-AI item 3 - Participants `:00571`	a) Describe how patients were recruited, stating the inclusion and exclusion criteria at both patient and data level, and how the number of recruited patients was decided.	sampling	R Ro Fa
DECIDE-AI item 4 - AI system `:00572`	a) Briefly describe the AI system, specifying its version and type of underlying algorithm used. Describe, or provide a direct reference to, the characteristics of the patient population on which the algorithm was trained and its performance in preclinical development/validation studies.	model architecture, published results identifier, version	I R Ex
DECIDE-AI item 5 - Implementation `:00573`	a) Describe the settings in which the AI system was evaluated.	clinical workflow phase	Un Us
DECIDE-AI item 6 - Safety and errors `:00575`	a) Provide a description of how significant errors/malfunctions were defined and identified.	known bias or failure mode	Ro Fa
DECIDE-AI item 7 - Human factors `:00576`	Describe the human factors tools, methods or frameworks used, the use cases considered, and the users involved.	—	Ex Us
DECIDE-AI item 8 - Ethics `:00578`	Describe whether specific methodologies were utilised to fulfil an ethics-related goal (such as algorithmic fairness) and their rationale.	ethical consideration	Fa
DECIDE-AI item 9 - Participants `:00580`	a) Describe the baseline characteristics of the patients included in the study, and report on input data missingness.	missing information, population	R Un Fa
DECIDE-AI item I - Abstract `:00567`	Provide a structured summary of the study. Consider including: intended use of the AI system, type of underlying algorithm, study setting, number of	summary	F Us

Guideline item	Definition	Metadata elements	Principles
	patients and users included, primary and secondary outcomes, key safety endpoints, human factors evaluated, main results, conclusions.		
DECIDE-AI item II - Objectives `:00569`	State the study objectives.	motivation	F
DECIDE-AI item III - Research governance `:00570`	Provide a reference to any study protocol, study registration number, and ethics approval.	clinical trial number, ethical review, reference	—
DECIDE-AI item IV - Outcomes `:00574`	Specify the primary and secondary outcomes measured.	—	Ro
DECIDE-AI item IX - Strengths and limitations `:00590`	Discuss the strengths and limitations of the study.	limitations	Un Ex
DECIDE-AI item V - Analysis `:00577`	Describe the statistical methods by which the primary and secondary outcomes were analysed, as well as any prespecified additional analyses, including subgroup analyses and their rationale.	—	Ro
DECIDE-AI item VI - Patient involvement `:00579`	State how patients were involved in any aspect of: the development of the research question, the study design, and the conduct of the study.	—	Us Fa
DECIDE-AI item VII - Main results `:00582`	Report on the prespecified outcomes, including outcomes for any comparison group if applicable.	model performance	Ex Ro
DECIDE-AI item VIII - Subgroups analysis `:00583`	Report on the differences in the main outcomes according to the prespecified subgroups.	—	Ex Fa
DECIDE-AI item X - Conflicts of interest `:00592`	Disclose any relevant conflicts of interest, including the source of funding for the study, the role of funders, any other roles played by commercial companies, and personal conflicts of interest for each author.	funding	—
MI-CLAIM item 01 `:00593`	The clinical problem in which the model will be employed is clearly detailed in the paper.	indication for use, model use	R Ex Us

Guideline item	Definition	Metadata elements	Principles
MI-CLAIM item 02 `:00594`	The research question is clearly stated.	motivation	F
MI-CLAIM item 03 `:00595`	The characteristics of the cohorts (training and test sets) are detailed in the text.	population, sampling	R Fa
MI-CLAIM item 04 `:00596`	The cohorts (training and test sets) are shown to be representative of real-world clinical settings.	known data-characterization gap, population	Un Ro Fa
MI-CLAIM item 05 `:00597`	The state-of-the-art solution used as a baseline for comparison has been identified and detailed.	—	R
MI-CLAIM item 06 `:00598`	The origin of the data is described and the original format is detailed in the paper.	external data	F A R
MI-CLAIM item 07 `:00599`	Transformations of the data before it is applied to the proposed model are described.	—	I R
MI-CLAIM item 08 `:00600`	The independence between training and test sets has been proven in the paper.	partitioning scheme, relationships between instances	Ro
MI-CLAIM item 09 `:00601`	Details on the models that were evaluated and the code developed to select the best model are provided.	model architecture, reproducibility	F A R Ex
MI-CLAIM item 10 `:00602`	Is the input data type structured or unstructured?	model input	I
MI-CLAIM item 11 `:00603`	The primary metric selected to evaluate algorithm performance (e.g., AUC, F-score, etc.), including the justification for selection, has been clearly stated.	model performance	Ex Ro
MI-CLAIM item 12 `:00604`	The primary metric selected to evaluate the clinical utility of the model (e.g., PPV, NNT, etc.), including the justification for selection, has been clearly stated.	clinical benefit	Ex Us
MI-CLAIM item 13 `:00605`	The performance comparison between baseline and proposed model is presented with the appropriate statistical significance.	—	Ro
MI-CLAIM item 14 `:00606`	Examination technique 1. Common examination approaches based on study type: for studies involving exclusively structured data, coefficients	—	Ex Ro

Guideline item	Definition	Metadata elements	Principles
	and sensitivity analysis are often appropriate; for studies involving unstructured data in the domains of image analysis or natural language processing, saliency maps (or equivalents) and sensitivity analyses are often appropriate.		
MI-CLAIM item 15 `:00607`	Examination technique 2. Common examination approaches based on study type: for studies involving exclusively structured data, coefficients and sensitivity analysis are often appropriate; for studies involving unstructured data in the domains of image analysis or natural language processing, saliency maps (or equivalents) and sensitivity analyses are often appropriate.	—	Ex Ro
MI-CLAIM item 16 `:00608`	A discussion of the relevance of the examination results with respect to model/algorithm performance is presented.	—	Ex Us
MI-CLAIM item 17 `:00609`	A discussion of the feasibility and significance of model interpretability at the case level if examination methods are uninterpretable is presented.	—	Ex Us
MI-CLAIM item 18 `:00610`	A discussion of the reliability and robustness of the model as the underlying data distribution shifts is included.	known bias or failure mode, limitations	Ex Ro
MI-CLAIM item 19 `:00611`	Reproducibility: choose appropriate tier of transparency.	license, model availability, reproducibility	R Ex Us
MINIMAR item 1.1 - Population `:00542`	Population from which study sample was drawn.	known data-characterization gap, population	R Un Fa
MINIMAR item 1.2 - Study setting `:00543`	The setting in which the study was conducted (eg, academic medical left, community healthcare system, rural healthcare clinic).	population	Un
MINIMAR item 1.3 - Data source `:00544`	The source from which data were collected.	external data, sampling	F A R
MINIMAR item 1.4 - Cohort selection `:00545`	Exclusion/inclusion criteria.	sampling	R Ro Fa

Guideline item	Definition	Metadata elements	Principles
MINIMAR item 2.1 - Age `:00547`	Age of patients included in the study.	population	Un Fa
MINIMAR item 2.2 - Sex `:00548`	Sex breakdown of study cohort.	population	Un Fa
MINIMAR item 2.3 - Race `:00549`	Race characteristics of patients included in the study.	population	Un Fa
MINIMAR item 2.4 - Ethnicity `:00550`	Ethnicity breakdown of patients included in the study.	population	Un Fa
MINIMAR item 2.5 - Socioeconomic status `:00551`	A measure or proxy measure of the socioeconomic status of patients included in the study.	population	Un Fa
MINIMAR item 3.1 - Model output `:00553`	The computed result of the model.	model output	Ex Us
MINIMAR item 3.2 - Target user `:00554`	The indented user of the model output (eg, clinician, hospital management team, insurance company).	model user	Ex Us
MINIMAR item 3.3 - Data splitting `:00555`	How data were split for training, testing, and validation.	partitioning scheme	R Ro
MINIMAR item 3.4 - Gold standard `:00556`	Labeled data used to train and test the model.	annotation level	R Ex
MINIMAR item 3.5 - Model task `:00557`	Classification or prediction.	model output	Ex
MINIMAR item 3.6 - Model architecture `:00558`	Algorithm type (eg, machine learning, deep learning, etc.).	model architecture	I Ex
MINIMAR item 3.7 - Features `:00559`	List of variables used in the model and how they were used in the model in terms of categories or transformation.	model input	I R Ex
MINIMAR item 3.8 - Missingness `:00560`	How missingness was addressed: reported, imputed, or corrected.	missing information	Ro

Guideline item	Definition	Metadata elements	Principles
MINIMAR item 4.1 - Optimization `:00562`	Model or parameter tuning applied.	—	Ex Ro
MINIMAR item 4.2 - Internal model validation `:00563`	Study internal validation.	partitioning scheme	Ex Ro
MINIMAR item 4.3 - External model validation `:00564`	External validation using data from another setting.	partitioning scheme	Un Ro
MINIMAR item 4.4 - Transparency `:00565`	How code and data are shared with the community..	license, link, model availability, reproducibility	F A R
PROBAST+AI model development item 1.1 `:00614`	Were appropriate data sources used?	external data, sampling	F A R
PROBAST+AI model development item 1.2 `:00615`	Was an appropriate study design used?	sampling strategy	Ro
PROBAST+AI model development item 1.3 `:00616`	Did the in- and exclusions of study participants result in a representative dataset?	known data-characterization gap, sampling	Un Ro Fa
PROBAST+AI model development item 2.1 `:00617`	Were predictors defined and assessed in a similar way for all participants?	model input	I R Ro
PROBAST+AI model development item 2.2 `:00618`	Was any pre-processing of predictors similar for all participants?	noise	I Ro
PROBAST+AI model development item 2.3 `:00619`	Were predictor assessments made without knowledge of outcome data?	—	Ro

Guideline item	Definition	Metadata elements	Principles
PROBAST+AI model development item 2.4 `:00620`	Were the predictors included in the model available at the time the model was intended to be used?	model input	Us Ro
PROBAST+AI model development item 3.1 `:00621`	Were outcomes defined and assessed appropriately?	annotation level	Ex Ro
PROBAST+AI model development item 3.2 `:00622`	Were outcomes defined and assessed in a similar way for all participants?	annotation level	Ro Fa
PROBAST+AI model development item 3.3 `:00623`	Were outcome assessments made without use or knowledge of predictor data?	—	Ro
PROBAST+AI model development item 3.4 `:00624`	Was the time interval between predictor assessment and outcome assessment appropriate?	—	Ro
PROBAST+AI model development item 4.1 `:00625`	Was there evidence that the sample size was reasonable?	sampling	Ro
PROBAST+AI model development item 4.2 `:00626`	Were continuous and categorical predictors handled appropriately?	model input	Ro
PROBAST+AI model development item 4.3 `:00627`	Were participants with missing or censored data handled appropriately in the analysis?	missing information	Ro
PROBAST+AI model development item 4.4 `:00628`	If methods to address class imbalance were used, was the model or the model predictions recalibrated?*	—	Ro Fa
PROBAST+AI model	Were methods used to address potential model overfitting?	partitioning scheme	Ex Ro

Guideline item	Definition	Metadata elements	Principles
development item 4.5 :00635			
PROBAST+AI model evaluation item 1.1 :00636	Were appropriate data sources used?	external data, sampling	F A R
PROBAST+AI model evaluation item 1.2 :00637	Was an appropriate study design used?	sampling strategy	Ro
PROBAST+AI model evaluation item 1.3 :00638	Did the in- and exclusions of study participants result in a representative dataset?	known data-characterization gap, sampling	Un Ro Fa
PROBAST+AI model evaluation item 2.1 :00639	Were predictors defined and assessed in a similar way for all participants?	model input	I R Ro
PROBAST+AI model evaluation item 2.2 :00640	Was any pre-processing of predictors similar for all participants?	noise	I Ro
PROBAST+AI model evaluation item 2.3 :00641	Were predictor assessments made without knowledge of outcome data?	—	Ro
PROBAST+AI model evaluation item 2.4 :00642	Were the predictors included in the model available at the time the model was intended to be used?	model input	Us Ro
PROBAST+AI model evaluation item 3.1 :00643	Were outcomes defined and assessed appropriately?	annotation level	Ex Ro
PROBAST+AI model evaluation item	Were outcomes defined and assessed in a similar way for all participants?	annotation level	Ro Fa

Guideline item	Definition	Metadata elements	Principles
3.2 :00644			
PROBAST+AI model evaluation item 3.3 :00645	Were outcome assessments made without use or knowledge of predictor data?	—	Ro
PROBAST+AI model evaluation item 3.4 :00646	Was the time interval between predictor assessment and outcome assessment appropriate?	—	Ro
PROBAST+AI model evaluation item 4.1 :00647	Was model evaluation based on only apparent performance avoided?	partitioning scheme	Ex Ro
PROBAST+AI model evaluation item 4.2 :00648	Was there evidence that the sample size was reasonable?	sampling	Ro
PROBAST+AI model evaluation item 4.3 :00649	Were participants with missing or censored data handled appropriately in the analysis?	missing information	Ro
PROBAST+AI model evaluation item 4.4 :00650	If methods to address class imbalance were used, was the evaluation done in a dataset without imbalance correction?*	—	Ro Fa
PROBAST+AI model evaluation item 4.5 :00651	If data splitting was done to create training and test datasets, was there evidence that data leakage was avoided?*	partitioning scheme, relationships between instances	Ro
PROBAST+AI model evaluation item 4.6 :00652	If resampling methods were used to evaluate model performance, were all model development steps replicated in the resampling process?*	partitioning scheme	R Ro
PROBAST+AI model evaluation item 4.7 :00653	Was the predictive performance of the model evaluated appropriately, e.g., calibration, discrimination, and net benefit?	model performance	Ex Ro

Guideline item	Definition	Metadata elements	Principles
REFINE item 1.1 `:00497`	Model name, vendor/developer, version/identifier, release date, and training/knowledge cutoff date.	date, name, version	F
REFINE item 1.2 `:00498`	Model architecture and key characteristics.	model architecture	I Ex
REFINE item 1.3 `:00499`	Model pretraining, post-training, and inference-time adaptation strategy.	model architecture	Ex Ro
REFINE item 1.4 `:00500`	Modality support details (input and output) and limitations.	limitations, model input, model output	I Ex Us
REFINE item 1.5 `:00501`	Language capabilities.	—	I Un Us
REFINE item 1.6 `:00502`	Model access.	link, model availability	A
REFINE item 1.7 `:00503`	Sharing of code, data, and model artifacts.	license, link, model availability	F A R
REFINE item 1.8 `:00504`	Computational requirements.	required processors, sustainability, time to evaluate single input, time to train model	Ro
REFINE item 2.1 `:00505`	Prompt engineering protocol with versioning.	instructions for use	Ro
REFINE item 2.2 `:00506`	Prompting strategy, format, and length.	instructions for use	I Ex
REFINE item 2.3 `:00507`	Prompt modality, language, technical input specification, and full content.	model input	I R
REFINE item 2.4 `:00508`	Integration of relevant patient clinical context.	—	Ex Us
REFINE item 2.5 `:00509`	Interaction style and session memory policy.	instructions for use	Ex Us
REFINE item 2.6 `:00510`	Output handling.	model output	Ex Us
REFINE item 3.1 `:00511`	Generation parameters.	decision threshold, reproducibility	Ro
REFINE item 3.2 `:00512`	Prompt operator characteristics and number of prompt attempts.	—	Ro
REFINE item 3.3 `:00513`	Output selection.	—	Ex
REFINE item 4.1 `:00514`	Dataset name, version, access type, source citation, license, and compliance statement.	license, published results identifier	F A R

Guideline item	Definition	Metadata elements	Principles
REFINE item 4.10 `:00523`	Separation of training, fine-tuning, internal testing, and external testing datasets.	partitioning scheme, relationships between instances	R Ro
REFINE item 4.2 `:00515`	Dataset origin.	external data, sampling	F A R
REFINE item 4.3 `:00516`	Ethics and consent statements.	confidentiality, ethical consideration, ethical review, risk of re-identification, sensitive data	—
REFINE item 4.4 `:00517`	Prior dataset usage and publication date.	external data, publication date	F
REFINE item 4.5 `:00518`	Dataset composition and data synthesis details.	sampling	I R
REFINE item 4.6 `:00519`	Sample characteristics and representational bias analysis.	known data-characterization gap, population	Un Ro Fa
REFINE item 4.7 `:00520`	Reference standard and annotator qualifications.	annotation level	R Ex
REFINE item 4.8 `:00521`	Preprocessing and data pairing/registration.	noise, relationships between instances	I R
REFINE item 4.9 `:00522`	Missing data extent, mechanism, and handling.	missing information	Ro
REFINE item 5.1 `:00524`	Output evaluation method and performance metrics.	model performance	Ex Ro
REFINE item 5.10 `:00533`	Comparison with clinically relevant benchmarks.	—	Ex Ro
REFINE item 5.2 `:00525`	Human evaluator characteristics and reliability analysis.	—	Ex Ro
REFINE item 5.3 `:00526`	Statistical analysis of evaluation results.	model performance	Ex Ro
REFINE item 5.4 `:00527`	Subgroup performance and output bias assessment.	known bias or failure mode	Ex Ro Fa
REFINE item 5.5 `:00528`	Failure analysis and error metrics.	known bias or failure mode	Ex Ro Fa
REFINE item 5.6 `:00529`	Output stochasticity and reproducibility constraints.	reproducibility	R Ex Ro
REFINE item 5.7 `:00530`	Performance effects of different prompt strategies and revisions.	—	Ex Ro

Guideline item	Definition	Metadata elements	Principles
REFINE item 5.8 `:00531`	Model version comparisons and temporal performance variation.	update date, version	Ro
REFINE item 5.9 `:00532`	Methods for explainability and interpretability of model outputs.	—	Ex Ro
REFINE item 6.1 `:00534`	Declared intended application and scope of use.	indication for use, model use, recommendation	Ex Us
REFINE item 6.2 `:00535`	Clinical workflow integration.	clinical workflow phase, degree of automation, instructions for use	Us
REFINE item 6.3 `:00536`	Measured clinical utility or added value.	clinical benefit	Ex Us
REFINE item 6.4 `:00537`	Model limitations, explicit clinical non-use cases, and potential misuse considerations.	excluded model use, limitations, out-of-scope model use	Un Ex
REFINE item 6.5 `:00538`	Safety testing and monitoring protocols.	known bias or failure mode	Ro Fa
REFINE item 6.6 `:00539`	Data security and privacy safeguards.	confidentiality, risk of re-identification, sensitive data	—
REFINE item 6.7 `:00540`	Governance, auditability, and oversight.	ethical consideration, regulatory information	Ex
SPIRIT-AI item 01 `:00345`	Descriptive title identifying study design, population, interventions, and trial acronym. Indicate intervention involves AI/ML and specify model type. Specify intended use of AI intervention.	model architecture, model use	F
SPIRIT-AI item 02a `:00370`	Trial identifier and registry name.	clinical trial number	F
SPIRIT-AI item 02b `:00371`	WHO Trial Registration Dataset items.	—	F
SPIRIT-AI item 03 `:00372`	Date and version identifier.	date, version	F
SPIRIT-AI item 04 `:00373`	Sources and types of support.	funding	—

Guideline item	Definition	Metadata elements	Principles
SPIRIT-AI item 05a :00374	Names, affiliations, roles of contributors.	contact information	—
SPIRIT-AI item 05b :00375	Trial sponsor contact info.	contact information	—
SPIRIT-AI item 05c :00376	Role of sponsor and funders.	funding	—
SPIRIT-AI item 06a :00378	Research question and justification. Explain intended AI use in clinical pathway. Describe pre-existing evidence.	model use, motivation	R Us
SPIRIT-AI item 06b :00379	Explanation for comparators.	—	Ex
SPIRIT-AI item 07 :00380	Objectives or hypotheses.	motivation	—
SPIRIT-AI item 08 :00381	Trial design description.	—	Ro
SPIRIT-AI item 09 :00382	Study settings and locations. Integration requirements for AI.	clinical workflow phase	Un Us
SPIRIT-AI item 10 :00383	Inclusion/exclusion criteria. Participant-level criteria. Input data criteria.	model input, sampling	R Ro Fa
SPIRIT-AI item 11a :00384	Intervention details. AI algorithm version. Input data acquisition. Handling poor-quality data. Human-AI interaction. Specify AI output. Use of AI output in decisions.	degree of automation, instructions for use, model input, model output, model user, version	I R Ex
SPIRIT-AI item 11b :00385	Criteria for discontinuation.	—	Ro
SPIRIT-AI item 11c :00386	Adherence strategies.	—	Ro
SPIRIT-AI item 11d :00387	Concomitant care.	—	Us

Guideline item	Definition	Metadata elements	Principles
SPIRIT-AI item 12 :00388	Primary and secondary outcomes.	—	Ro
SPIRIT-AI item 13 :00389	Schedule of enrollment and visits.	—	Ro
SPIRIT-AI item 14 :00390	Sample size determination.	—	Ro
SPIRIT-AI item 15 :00391	Recruitment strategies.	—	Un Fa
SPIRIT-AI item 16a :00392	Allocation sequence generation.	—	Ro
SPIRIT-AI item 16b :00393	Concealment mechanism.	—	Ro
SPIRIT-AI item 17a :00395	Blinding details.	—	Ro
SPIRIT-AI item 17b :00396	Unblinding conditions.	—	Ro
SPIRIT-AI item 18a :00397	Outcome and baseline data collection.	—	R Ro
SPIRIT-AI item 18b :00398	Retention and follow-up.	—	Ro
SPIRIT-AI item 19 :00399	Data entry and storage.	confidentiality	A I
SPIRIT-AI item 20a :00400	Analysis of outcomes.	model performance	Ex Ro
SPIRIT-AI item 20b :00401	Additional analyses.	—	Ex Fa
SPIRIT-AI item 20c :00402	Missing data handling.	missing information	Ro

Guideline item	Definition	Metadata elements	Principles
SPIRIT-AI item 21b `:00404`	Interim analyses.	—	Ro
SPIRIT-AI item 22 `:00405`	Adverse events handling. AI performance errors analysis.	known bias or failure mode	Ex Ro Fa
SPIRIT-AI item 24 `:00407`	IRB approval plans.	ethical review	—
SPIRIT-AI item 25 `:00408`	Communication of changes.	—	Us
SPIRIT-AI item 26a `:00409`	Informed consent process.	ethical review	—
SPIRIT-AI item 27 `:00411`	Participant data protection.	confidentiality, risk of re-identification, sensitive data	—
SPIRIT-AI item 29 `:00413`	Dataset access. Access to AI code.	license, link, model availability	F A R
SPIRIT-AI item 30 `:00414`	Post-trial care.	—	Us
SPIRIT-AI item 31a `:00415`	Communication of results.	—	F
SPIRIT-AI item 31c `:00417`	Public access to protocol/data.	—	F A
SPIRIT-AI item 33 `:00419`	Specimen handling.	—	R
STARD-AI item 01 `:00279`	Identification as a study reporting AI-centered diagnostic accuracy and reporting at least one measure of accuracy within title or abstract.	model performance	F
STARD-AI item 02 `:00280`	Structured summary of study design, methods, results and conclusions (for specific guidance, please see STARD for Abstracts).	summary	F
STARD-AI item 03 `:00281`	Scientific and clinical background, including the intended use of the index test, whether it is novel or an	clinical workflow phase, indication for	R Us

Guideline item	Definition	Metadata elements	Principles
	established index test, and its integration into an existing or new workflow, if applicable	use, model use, motivation	
STARD-AI item 04 `:00282`	Study objectives and hypotheses.	motivation	F
STARD-AI item 05 `:00283`	Whether data collection was planned before the index test and reference standard were performed (prospective study) or after (retrospective study).	—	Ro
STARD-AI item 06 `:00284`	Formal approval from an ethics committee. If not required, justify why.	ethical review	—
STARD-AI item 07 `:00285`	Eligibility criteria: listing separate inclusion and exclusion criteria in the order that they are applied at both participant level and data level.	sampling	R Ro Fa
STARD-AI item 08 `:00286`	On what basis potentially eligible participants were identified (such as symptoms, results from previous tests, inclusion in registry).	sampling	R Un
STARD-AI item 09 `:00287`	Where and when potentially eligible participants were identified (setting, location, and dates).	sampling	R Un
STARD-AI item 10 `:00288`	Whether participants formed a consecutive, random, or convenience series.	sampling strategy	R Ro Fa
STARD-AI item 11 `:00289`	Source of the data and whether it has been routinely collected, specifically collected for the purpose of the study or acquired from an open-source repository.	external data, sampling	F A R
STARD-AI item 12 `:00290`	Who undertook the annotations for the dataset (including experience levels and background) and how (within the same clinical context or in a post-hoc fashion), if applicable.	annotation level	R Ex
STARD-AI item 13 `:00291`	Devices (manufacturer, model) that were used to capture data; software (with version number) used to engineer the index test, highlighting the intended use.	—	I Ro
STARD-AI item 14 `:00292`	Data acquisition protocols (e.g. contrast protocol or reconstruction method for medical images) and details of data	—	I R Ro

Guideline item	Definition	Metadata elements	Principles
	pre-processing in sufficient detail to allow replication.		
STARD-AI item 15a `:00293`	Index test, in sufficient detail to allow replication.	model input	I R Ex
STARD-AI item 15b `:00294`	How the index test was developed, including any training, validation, testing and external evaluation, detailing sample sizes, when applicable.	partitioning scheme	Ro
STARD-AI item 15c `:00295`	Definition of and rationale for test positivity cut-offs or result categories of the index test, distinguishing pre-specified from exploratory.	decision threshold	Ex Ro
STARD-AI item 15d `:00296`	The specified end user of the index test and the level of expertise required of users.	model user	Ex Us
STARD-AI item 16a `:00297`	Reference standard, in sufficient detail to allow replication.	annotation level	I R
STARD-AI item 16b `:00298`	Rationale for choosing the reference standard (if alternatives exist).	—	Ex
STARD-AI item 16c `:00299`	Definition of and rationale for test positivity cut-offs or result categories of the reference standard, distinguishing pre-specified from exploratory.	—	Ex
STARD-AI item 17a `:00300`	Whether clinical information and reference standard results were available to the performers or readers of the index test.	—	Ro
STARD-AI item 17b `:00301`	Whether clinical information and index test results were available to the assessors of the reference standard.	—	Ro
STARD-AI item 18 `:00302`	Methods for estimating or comparing measures of diagnostic accuracy.	model performance	R Ex
STARD-AI item 19 `:00303`	How indeterminate index test or reference standard results were handled.	missing information	Ro
STARD-AI item 20 `:00304`	How missing data on the index test and reference standard were handled.	missing information	Ro
STARD-AI item 21	Any analyses of variability in diagnostic accuracy, distinguishing	—	Ex Ro

Guideline item	Definition	Metadata elements	Principles
:00305	pre-specified from exploratory.		
STARD-AI item 22 :00306	Intended sample size and how it was determined.	sampling	Ro
STARD-AI item 23 :00307	Details of any performance error analysis, and algorithmic bias and fairness assessments if undertaken.	known bias or failure mode	Ex Ro Fa
STARD-AI item 25 :00309	Baseline demographic, clinical and technical characteristics of training, validation and test set, if applicable.	partitioning scheme, population	R Un Ro Fa
STARD-AI item 26a :00310	Distribution of severity of disease in those with the target condition.	—	R Un Fa
STARD-AI item 26b :00311	Distribution of alternative diagnoses in those without the target condition.	—	R Un Fa
STARD-AI item 27 :00312	Time interval and any clinical interventions between index test and reference standard.	—	Ro
STARD-AI item 28 :00313	Whether the datasets represent the distribution of the target condition that one would expect from the intended use population.	known data-characterization gap, population	Un Ro Fa
STARD-AI item 29 :00314	For external evaluation on an independent dataset, an assessment of how this differs from the training, validation and test sets.	known data-characterization gap	Un Ro
STARD-AI item 30 :00315	Cross tabulation of the index test results (or their distribution) by the results of the reference standard.	—	Ex
STARD-AI item 31 :00316	Estimates of diagnostic accuracy and their precision (such as 95% confidence intervals).	model performance	Ex Ro
STARD-AI item 32 :00317	Any adverse events from performing the index test or the reference standard.	—	Ro
STARD-AI item 33 :00318	Study limitations, including sources of potential bias, statistical uncertainty, and generalisability.	limitations, noise	Un Ex Ro
STARD-AI item 34 :00319	Implications for practice, including the intended use and clinical role of the index test.	clinical benefit, model use, recommendation	Ex Us

Guideline item	Definition	Metadata elements	Principles
STARD-AI item 35 :00320	Ethical considerations and adherence to ethical standards associated with the use of the index test and issues of fairness.	ethical consideration	Fa
STARD-AI item 36 :00321	Registration number and name of registry.	clinical trial number	F
STARD-AI item 37 :00322	Where the full study protocol can be accessed.	reference	F A
STARD-AI item 39 :00324	Commercial interests, if applicable.	funding	—
STARD-AI item 40a :00325	Availability of datasets and code; detailing any restrictions on their reuse and repurposing.	—	F A R
STARD-AI item 40b :00326	Whether outputs are stored, auditable and available for evaluation, if necessary.	—	A Ex
TRAM-AI item A1 :00180	Dataset is hosted on a recognized, third-party repository (e.g., Zenodo, TCIA, PhysioNet), not personal websites.	link	F A
TRAM-AI item A2 :00181	Dataset is assigned a DOI or stable accession number.	link	F A
TRAM-AI item A3 :00182	Repository supports explicit versioning; specific version cited in manuscript.	version	F A R
TRAM-AI item B1 :00184	Non-proprietary formats (DICOM, nifti). Preprocessing documented or raw data stated.	image file format, pre-processing, resolution	I R
TRAM-AI item B2 :00185	Cryptographic hashes (MD5/SHA-256) provided.	—	A I Ro
TRAM-AI item B3 :00186	Folder hierarchy documented.	—	I R
TRAM-AI item B4 :00187	Archives confirmed to extract on different OS.	—	I Ro
TRAM-AI item C1 :00189	Machine-readable file links Image identifiers to Subject identifiers.	—	I R

Guideline item	Definition	Metadata elements	Principles
TRAM-AI item C2 :00190	Training, validation, and test split IDs provided as static lists.	—	R Ro
TRAM-AI item C3 :00192	Labels and reference standard source defined.	—	R Fa
TRAM-AI item C4 :00193	Excluded IDs and reasons provided.	—	R Fa
TRAM-AI item D1 :00195	Standard license (CC-BY, CC0, ODbL, ODC-BY) applied.	license	A R
TRAM-AI item D2 :00196	PHI removal from headers and pixel data confirmed.	confidentiality, risk of re-identification, sensitive data	—
TRAM-AI item D3 :00197	Contact for error reporting; deprecation process established.	contact information	A R Us
TRIPOD+AI item 01 :00420	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted.	—	F
TRIPOD+AI item 02 :00421	See TRIPOD+AI for Abstracts checklist.	summary	F
TRIPOD+AI item 03a :00422	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models.	indication for use, motivation	R Ex Us
TRIPOD+AI item 03b :00423	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (e.g., healthcare professionals, patients, public).	clinical workflow phase, model use	R Us
TRIPOD+AI item 03c :00424	Describe any known health inequalities between sociodemographic groups.	known data-characterization gap	Un Fa
TRIPOD+AI item 04 :00425	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both).	motivation	F

Guideline item	Definition	Metadata elements	Principles
TRIPOD+AI item 05a `:00426`	Describe the sources of data separately for the development and evaluation datasets (e.g., randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data.	sampling, sampling strategy	F A R
TRIPOD+AI item 06a `:00428`	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including the number and location of centres.	—	Un
TRIPOD+AI item 06b `:00429`	Describe the eligibility criteria for study participants.	sampling	R Ro Fa
TRIPOD+AI item 06c `:00430`	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant.	—	R
TRIPOD+AI item 07 `:00431`	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups.	noise	I R Ro
TRIPOD+AI item 08a `:00432`	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups.	annotation level	Ex Ro
TRIPOD+AI item 08b `:00433`	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors.	—	R Ex
TRIPOD+AI item 08c `:00434`	Report any actions to blind assessment of the outcome to be predicted.	—	Ro
TRIPOD+AI item 09a `:00435`	Describe the choice of initial predictors (e.g., literature, previous models, all available predictors) and any pre-selection of predictors before model building.	—	R Ex
TRIPOD+AI item 09b `:00436`	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors).	model input	I R

Guideline item	Definition	Metadata elements	Principles
TRIPOD+AI item 09c `:00437`	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors.	—	R
TRIPOD+AI item 10 `:00438`	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation.	sampling	Ro
TRIPOD+AI item 11 `:00439`	Describe how missing data were handled. Provide reasons for omitting any data.	missing information	Ro
TRIPOD+AI item 12a `:00440`	Describe how the data were used (e.g., for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements.	partitioning scheme, relationships between instances	Ro
TRIPOD+AI item 12b `:00441`	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation).	—	I Ex
TRIPOD+AI item 12c `:00442`	Specify the type of model, rationale2, all model-building steps, including any hyperparameter tuning, and method for internal validation.	model architecture, partitioning scheme	I Ex Ro
TRIPOD+AI item 12d `:00443`	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (e.g., hospitals, countries). See TRIPOD-Cluster for additional considerations.	relationships between instances	Ex Ro
TRIPOD+AI item 12e `:00444`	Specify all measures and plots used (and their rationale) to evaluate model performance (e.g., discrimination, calibration, clinical utility) and, if relevant, to compare multiple models.	model performance	Ex Ro
TRIPOD+AI item 12f `:00445`	Describe any model updating (e.g., recalibration) arising from the model evaluation, either overall or for particular sociodemographic groups or settings.	—	Ex
TRIPOD+AI item 12g	For model evaluation, describe how the model predictions were calculated (e.g.,	—	I Ex

Guideline item	Definition	Metadata elements	Principles
:00446	formula, code, object, application programming interface).		
TRIPOD+AI item 13 :00447	If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions.	—	Ro Fa
TRIPOD+AI item 14 :00448	Describe any approaches that were used to address model fairness and their rationale.	ethical consideration	Ex Ro Fa
TRIPOD+AI item 15 :00449	Specify the output of the prediction model (e.g., probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified.	decision threshold, model output	I R Ex
TRIPOD+AI item 16 :00450	Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors.	known data-characterization gap	Un Ro
TRIPOD+AI item 17 :00451	Name the institutional research board or ethics committee that approved the study and describe the participant-informed consent or the ethics committee waiver of informed consent.	ethical review	—
TRIPOD+AI item 18a :00462	Give the source of funding and the role of the funders for the present study.	funding	—
TRIPOD+AI item 18b :00452	Declare any conflicts of interest and financial disclosures for all authors.	funding	—
TRIPOD+AI item 18c :00453	Indicate where the study protocol can be accessed or state that a protocol was not prepared.	reference	F A
TRIPOD+AI item 18d :00454	Provide registration information for the study, including register name and registration number, or state that the study was not registered.	clinical trial number	F
TRIPOD+AI item 18e :00455	Provide details of the availability of the study data.	link	F A R
TRIPOD+AI item 18f :00456	Provide details of the availability of the analytical code7.	link, model availability, reproducibility	F A R
TRIPOD+AI item 19 :00457	Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or	—	Us Fa

Guideline item	Definition	Metadata elements	Principles
	dissemination of the study or state no involvement.		
TRIPOD+AI item 20b `:00458`	Report the characteristics overall and, where applicable, for each data source or setting, including the key dates, key predictors (including demographics), treatments received, sample size, number of outcome events, follow-up time, and amount of missing data. A table may be helpful. Report any differences across key demographic groups.	population	R Un Fa
TRIPOD+AI item 20c `:00459`	For model evaluation, show a comparison with the development data of the distribution of important predictors (demographics, predictors, and outcome)..	known data-characterization gap	Un Ro Fa
TRIPOD+AI item 21 `:00460`	Specify the number of participants and outcome events in each analysis (e.g., for model development, hyperparameter tuning, model evaluation).	—	Ro
TRIPOD+AI item 22 `:00461`	Provide details of the full prediction model (e.g., formula, code, object, API) to allow predictions in new individuals and to enable third-party evaluation and implementation, including any restrictions to access or re-use (e.g., freely available, proprietary)8.	license, link, model availability	A I R Ex
TRIPOD+AI item 23a `:00464`	Report model performance estimates with confidence intervals, including for any key subgroups (e.g., sociodemographic). Consider plots to aid presentation.	model performance	Ex Ro Fa
TRIPOD+AI item 23b `:00465`	If examined, report results of any heterogeneity in model performance across clusters. See TRIPOD Cluster for additional details.	—	Ro Fa
TRIPOD+AI item 24 `:00466`	Report the results from any model updating, including the updated model and subsequent performance.	—	Ex
TRIPOD+AI item 25 `:00467`	Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies.	—	Ex Us Fa

Guideline item	Definition	Metadata elements	Principles
TRIPOD+AI item 26 `:00468`	Discuss any limitations of the study (such as a non-representative sample, sample size, overfitting, missing data) and their effects on any biases, statistical uncertainty, and generalizability.	limitations	Un Ro Fa
TRIPOD+AI item 27a `:00469`	Describe how poor quality or unavailable input data (e.g., predictor values) should be assessed and handled when implementing the prediction model.	instructions for use, missing information, model input	Us Ro
TRIPOD+AI item 27b `:00470`	Discuss whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users.	degree of automation, instructions for use, model user	Ex Us
TRIPOD+AI item 27c `:00471`	Discuss any next steps for future research, with a specific view to applicability and generalizability of the model.	—	Un Ex Ro
guideline item `:00177`	An item of an AI reporting guideline, such as CLAIM or TRAM-AI. Note that these guideline items generally apply to scientific manuscripts (journal articles), rather than the descriptions of the AI models or datasets themselves.	—	Tr

Audit counts

	Count
Guideline items in the table	374
— support a principle **and** are supported by metadata	237
— support a principle but **no metadata** points to them	118
— supported by metadata but support **no principle**	19
Metadata to guideline-item links	415
Guideline-item to principle links	709
Metadata to principle links (direct)	36